IE500 Data Mining — Project Report

Early Prediction of Diabetes Risks

Using Machine Learning on the CDC BRFSS 2023 Dataset

Muhammad Murodzoda   Fatima Iqbal   Özlem Ölçer
Tim Mauscherning   Manuel Dieterle   Xuemei Wang

May 2026

Contents

1 Application Area and Goals

Diabetes is one of the most prevalent chronic diseases worldwide, with a steadily increasing incidence. According to International Diabetes Federation (2025), approximately 589 million adults aged 20–79 are currently affected, and this number is projected to rise to 853 million by 2050. Similar trends are observed at the country level: in Germany, about 6.5 million adults (10–11%) are affected, compared to 38.5 million in the United States. In both countries, prevalence has increased markedly over time, rising by roughly 2.5 times since 2000 (International Diabetes Federation 2025). In addition, the age of onset has been decreasing, with growing incidence among younger adults aged 30–45 (American Diabetes Association 2024).

Given these developments, early identification of high-risk individuals is crucial. Evidence from large-scale randomized controlled trials shows that lifestyle interventions can reduce diabetes risk by 40–70% (Knowler et al. 2002; Tuomilehto et al. 2001). However, once diagnosed, diabetes typically requires long-term management and is associated with reduced quality of life due to symptoms such as fatigue, vision problems, and impaired wound healing (Rubin and Peyrot 1999; National Health Service 2023).

Against this background, this project aims to develop machine-learning–based predictive models to identify individuals at elevated risk of diabetes and pre-diabetes using demographic, behavioral, and health-related variables drawn from large-scale survey data. The goal is to build a data-driven screening tool that supports early detection and preventive health management, while systematically comparing model families with respect to their ability to handle class imbalance and identify key factors associated with diabetes risk.

2 Dataset Profile

2.1 Source and Size

The analysis is based on data from the 2023 Behavioral Risk Factor Surveillance System (BRFSS), an annual health survey conducted by the Centers for Disease Control and Prevention (Centers for Disease Control and Prevention 2024). The BRFSS collects state-level data on health-related risk behaviors, chronic health conditions, and demographic characteristics across the United States, and constitutes one of the largest ongoing population health surveys in the world. The data is publicly available and can be accessed directly from the CDC website.

The raw 2023 dataset, distributed in SAS transport format (.XPT), contains more than 430,000 observations. After data cleaning the dataset was reduced to approximately 259,000 observations suitable for modeling. The final cleaned dataset was partitioned into a training set (70%) and a held-out test set (30%, n = 77,781) using stratified sampling to preserve the class distribution across both splits.

2.2 Feature Overview

A subset of 27 predictive features and one target variable was selected based on established diabetes-related risk factors and prior research (Centers for Disease Control and Prevention 2024), spanning three dimensions: metabolic indicators (e.g., BMI, high blood pressure, high cholesterol), behavioral factors (e.g., smoking status, physical activity, alcohol consumption), and socioeconomic variables (e.g., age, income, education, healthcare access). The target variable, derived from BRFSS item *DIABETE4*, was binarized as 1 for respondents reporting a diagnosis of diabetes or pre-diabetes and 0 otherwise. A full list of selected variables and their descriptions is provided in Table 1 (Section 3.1). In addition, four composite features were engineered to capture higher-order clinical risk patterns: *Metabolic_Syndrome*, *High_CVD_Risk_Profile*, *Comorbidity_Count*, and *Healthy_Lifestyle_Score*, as detailed in Section 3.2. The final feature set comprises three continuous variables, eight ordinal variables including three of the four engineered composites, and sixteen binary variables including *High_CVD_Risk_Profile*.

2.3 Class Distribution and Imbalance

The dataset exhibits a substantial class imbalance. In the held-out test set, 11,045 of 77,781 respondents (approximately 14.2%) are labelled as having diabetes or prediabetes, with the remaining 85.8% classified as non-diabetic. This ratio reflects the real-world prevalence of diagnosed diabetes and pre-diabetes combined in the U.S. adult population (International Diabetes Federation 2025) and is consistent across the full cleaned dataset. The imbalance poses a direct modeling challenge: a naive classifier predicting the majority class at all times would achieve over 85% accuracy while failing entirely to identify at-risk individuals. This directly motivated the use of class-weighted loss functions, recall-oriented evaluation metrics, and sampling strategies explored throughout the project, as discussed in Sections 3.4 through 3.4 and Section 5.1.

3 Preprocessing and Data Mining

3.1 Data Cleaning

The data cleaning process ensures consistency, removes noise, and transforms the raw survey data into a structured format suitable for machine learning. The CDC Behavioral Risk Factor Surveillance System (BRFSS) 2023 dataset was extracted and loaded into a pandas DataFrame with appropriate encoding. A subset of relevant features was selected based on established Type 2 diabetes risk factors spanning metabolic, behavioral, and socioeconomic dimensions, in line with CDC guidelines (Centers for Disease Control and Prevention 2024), and is summarized in Table 1.

Duplicate records were identified and removed using a composite key of *SEQNO*, *QSTVER*, *QSTLANG*, and *_STATE* which uniquely identifies each respondent within a given survey version, language, and state. Rows with missing or invalid values were dropped entirely.

Feature (Original)	Renamed	Description
DIABETE4	Diabetes	Diabetes status (target variable)
_RFHYPE6	HighBP	High blood pressure indicator
TOLDHI3	HighChol	High cholesterol diagnosis
_CHOLCH3	CholCheck	Cholesterol check status
_BMI5	BMI	Body Mass Index
_RFBMI5	Overweight	Overweight/obesity indicator
_SMOKER3	SmokerStatus	Smoking status (ordinal)
CVDSTRK3	Stroke	History of stroke
_MICHD	HeartDisease	Coronary heart disease indicator
_TOTINDA	PhysActivity	Physical activity indicator
_RFDRHV8	HeavyAlcohol	Heavy alcohol consumption
_HLTHPL1	HealthInsurance	Health insurance coverage
MEDCOST1	NoDocCost	Could not afford doctor visit
GENHLTH	GenHealth	General health status (ordinal)
MENTHLTH	MentHealth	Days of poor mental health
PHYSHLTH	PhysHealth	Days of poor physical health
DIFFWALK	DiffWalking	Difficulty walking
SEXVAR	Sex	Gender
_AGEG5YR	AgeGroup	Age group (ordinal)
EDUCA	Education	Education level
INCOME3	Income	Income level
ADDEPEV3	Depression	Depression diagnosis
HAVARTH4	Arthritis	Arthritis diagnosis
CHCKDNY2	KidneyDisease	Kidney disease diagnosis

Table 1: Selected variables and their descriptions. Metadata fields (*SequenceNo*, *PSU*, *State*, *QuestionnaireVersion*, *QuestionnaireLang*) were used for deduplication but excluded from modeling.

Each feature was recoded to produce clean binary or ordinal representations following BRFSS coding conventions. For binary features, "No" responses (coded as 2) were remapped to 0, and ambiguous entries such as "Don't know" (7) and "Refused" (9) were treated as invalid and removed, as they are not meaningful in a clinical context. In *MENTHLTH* and *PHYSHLTH*, the "None" code (88) was remapped to 0 before removing invalid entries (77, 99). As noted in Section 2, the target variable *DIABETE4* was binarized as 1 for diabetes or pre-diabetes and 0 otherwise, with invalid responses removed. BMI values were rescaled by dividing by 100 to recover the standard scale.

The BRFSS dataset contains no free-text fields, as all variables are pre-coded, and no cross-variable consistency checks were needed due to the survey's built-in validation. Invalid entries in age (*_AGEG5YR*), education (*EDUCA*), and income (*INCOME3*) were removed. After cleaning, variables were renamed to more interpretable labels. Cross-variable checks were unnecessary since age is recorded in predefined groups, and text normalization was not required due to numeric encoding. Continuous variables *BMI*, *MentHealth*, *PhysHealth* were kept in their original form and standardized during modeling.

3.2 Feature Engineering

Four composite variables were constructed to capture higher-order clinical risk patterns not directly observable from individual survey items. *Metabolic_Syndrome* is an ordinal proxy score (0–3) summing the binary indicators for obesity, hypertension, and high cholesterol, representing the cumulative burden of these jointly increasing the risk of type 2 diabetes, as defined in Equation (1):

$$Metabolic\_Syndrome = \mathbf{1}[Overweight] + \mathbf{1}[HighBP] + \mathbf{1}[HighChol] \tag{1}$$

The high cardiovascular risk profile *High_CVD_Risk_Profile* feature is a binary indicator flagging respondents with an elevated cardiovascular risk profile: a history of heart disease or stroke, or the co-occurrence of hypertension, high cholesterol, and current smoking as in Equation (2). This combination represents a clinically recognized pattern strongly associated with insulin resistance and metabolic disorders.

$$\begin{aligned} High\_CVD\_Risk\_Profile = \mathbf{1}[&(HeartDisease{=}1) \lor (Stroke{=}1) \\ &\lor (HighBP{=}1 \land HighChol{=}1 \land CurrentSmoker{=}1)] \end{aligned} \tag{2}$$

Comorbidity_Count is an additive count of three organ-level disease indicators; coronary heart disease, stroke, and kidney disease as shown in Equation (3). It serves as a simplified proxy for the Charlson Comorbidity Index (CCI), capturing the cumulative physiological stress that correlates with metabolic dysfunction.

$$Comorbidity\_Count = \mathbf{1}[HeartDisease] + \mathbf{1}[Stroke] + \mathbf{1}[KidneyDisease] \tag{3}$$

A *Healthy_Lifestyle_Score* is an ordinal score (0–4) summing four protective behavior indicators: non-smoking, physical activity, normal BMI, and no heavy alcohol use as in Equation (4). It captures the combined predictive signal of lifestyle factors that may be weak individually but informative together.

$$\begin{aligned} Healthy\_Lifestyle\_Score = \mathbf{1}[NonSmoker] + \mathbf{1}[PhysActivity] \\ + \mathbf{1}[\neg Overweight] + \mathbf{1}[\neg HeavyAlcohol] \end{aligned} \tag{4}$$

The final feature set comprises three continuous variables (*BMI*, *PhysHealth*, *MentHealth*) that were standardized, eight ordinal variables including the four engineered composites, and sixteen binary variables. Metadata fields were excluded as non-predictive.

3.3 Exploratory Data Analysis

Following data cleaning and feature engineering, an exploratory data analysis was conducted to understand the distributions within the processed dataset. Boxplots were generated for the three continuous features BMI, MentHealth, and PhysHealth (Figure 1). BMI shows a right-skewed distribution with high-end outliers, consistent with the known association between obesity and diabetes risk. Both health day variables are heavily right-skewed, with most respondents reporting zero poor health days. These distributions confirm that standardisation is appropriate for these features within the modelling pipeline. The pairwise correlation analysis revealed that GenHealth shows the strongest positive correlation with the target variable, followed by BMI, HighBP, and HighChol, informing the final feature selection for model training.

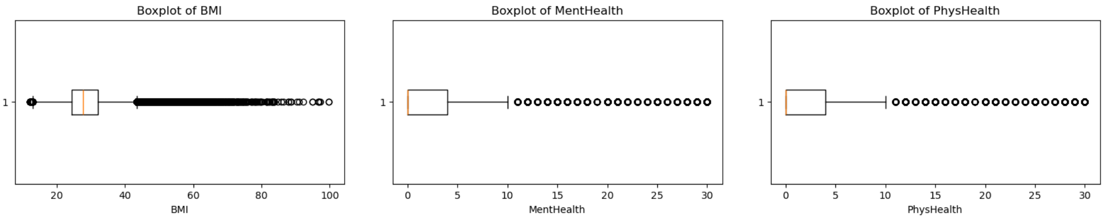

Figure 1: Boxplots for the three continuous features: BMI (left), MentHealth (center), and PhysHealth (right).

3.4 Modeling Approaches

To ensure a thorough comparison, ten classification models were selected covering a broad range of algorithmic families. The selection was deliberately diverse, moving from simple interpretable models through to more complex ensemble methods, allowing us to assess whether added complexity translates into meaningful performance gains on this dataset.

Logistic Regression serves as the linear baseline with balanced class weighting applied to prevent the model from defaulting toward the majority class. Gaussian Naive Bayes and Complement Naive Bayes provide probabilistic baselines, with the latter requiring min-max scaling due to its non-negative input constraint. A Support Vector Machine via stochastic gradient descent was also included for its scalability to large datasets.

The remaining models are tree-based ensembles. XGBoost, CatBoost, and LightGBM are sequential gradient boosting frameworks with positive class weighting that are tuned to address class imbalance. AdaBoost and Histogram Gradient Boosting provide additional boosting baselines, and Balanced Random Forest handles class imbalance through equal-class bootstrap sampling rather than class weights. All models are embedded in a preprocessing pipeline, ensuring scaling is applied strictly within each cross-validation fold to prevent data leakage.

Nested Cross-Validation Set-Up

The dataset was split into a training set (70%) and a held-out test set (30%) using stratified sampling to preserve the class ratio. The test set is set aside entirely and only used for final evaluation after model selection is complete. All hyperparameter tuning and model selection is performed exclusively on the training set using nested cross-validation, consisting of a 4-fold stratified inner loop via GridSearchCV optimising on

F1-score, wrapped by a 4-fold stratified outer loop evaluating generalisation performance. This separation of tuning and evaluation avoids the optimistic bias that arises when the same data is used for both.

Used Metrics

Model performance is reported across five metrics: Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Given the class imbalance, accuracy alone is insufficient, and Recall is of particular importance in this context, as failing to identify a diabetic patient carries a higher cost than a false alarm. F1-Score balances precision and recall and serves as the inner loop optimisation criterion, while AUC-ROC captures discriminative ability across all classification thresholds independently of class distribution.

4 Evaluation

4.1 Results of the Nested Cross-Validation

Following the nested cross-validation, two distinct groups emerge (Table 2): gradient-boosted models (LightGBM, XGBoost, CatBoost) that achieve the best F1-score and AUC-ROC, and class-balanced classifiers (Logistic Regression, SVM, Balanced RF) that sacrifice some F1 for substantially higher recall.. AdaBoost and Gradient Boosting achieve the highest accuracy and precision, but their recall drops below 0.20, meaning they fail to detect over 80% of actual diabetic cases. In a diagnostic context where missing a positive case carries higher cost than a false alarm, these models offer little practical utility.

LightGBM, CatBoost, and XGBoost achieve the highest F1-scores ($\sim$0.49) and AUC-ROC of 0.81, maintaining a balanced trade-off between precision ($\sim$0.40) and recall (0.62). Logistic Regression, SVM, and Balanced Random Forest yield slightly lower F1-scores ($\sim$0.47) but considerably higher recall (0.73–0.77), making them the least likely to produce false negatives. Naive Bayes models perform competitively given their simplicity and remain a viable baseline. All models show standard deviations ≤ 0.02 across folds, confirming high stability.

4.2 Hyperparameter Optimization and Stability

Following the nested cross-validation, an optimization process was conducted to determine the final hyperparameters for the models. Using an additional 5-fold cross-validation process, the models were optimized specifically for the F1-score. Consistent with the patterns observed during the nested cross-validation, the highest F1-scores were achieved by the XGBoost (0.4918) and LightGBM (0.4919) models. Generally, the F1-scores of the optimized models deviated only marginally, if at all, from their average performances in the preceding steps. This strongly suggests that the models are robust and stable across different data splits. To confirm this, an additional sanity check was performed: the final, optimized models were subjected to another 5-fold cross-validation

Model	Acc.	F1	Recall	Prec.	AUC
LightGBM	0.780	**0.491**	0.622	0.405	0.810
XGBoost	0.780	0.490	0.621	0.405	**0.810**
CatBoost	0.780	0.490	0.618	0.405	0.809
Logistic Regression	0.719	0.475	0.745	0.349	0.804
SVM	0.722	0.474	0.733	0.351	0.802
Balanced RF	0.705	0.471	**0.769**	0.340	0.805
NB Complement	0.732	0.444	0.629	0.344	0.769
NB Gaussian	0.763	0.441	0.548	0.370	0.772
AdaBoost	0.838	0.313	0.216	0.571	0.806
Gradient Boosting	**0.839**	0.297	0.199	**0.586**	0.807

Table 2: Nested CV performance (4-fold outer/inner; all std. ≤ 0.02; majority-class baseline $\approx$85% accuracy).

to verify the stability of all other evaluation metrics. The resulting trend remained completely consistent, confirming the models' high overall stability.

Ultimately, within the training data, the LightGBM model performs best in terms of the F1-score, while the Balanced Random Forest model yields the highest recall. XGBoost and LightGBM are selected as the preferred models, with the Balanced Random Forest as an alternative that minimizes false negative diagnoses.

Model	Key Hyperparameters
LightGBM	lr=0.1, depth=3, n_est=200, scale_pos_weight=3
XGBoost	lr=0.3, depth=3, n_est=100, scale_pos_weight=3
CatBoost	depth=4, iterations=500, scale_pos_weight=3
Balanced RF	depth=10, n_est=500
Logistic Regression	C=0.01, class_weight=balanced, solver=saga
SVM (SGD)	alpha=0.0001, loss=log_loss, class_weight=balanced
Gaussian NB	var_smoothing=10^{-5}
Complement NB	alpha=10 (MinMaxScaler)
AdaBoost	lr=1, n_est=50
Gradient Boosting	lr=0.3, depth=10

Table 3: Final hyperparameters (5-fold GridSearchCV, optimizing F1).

5 Results

5.1 Ensemble and Resampling Experiments

Soft and hard voting ensembles were constructed from six models (Balanced RF, SVM, LightGBM, CatBoost, Logistic Regression, XGBoost). As Table 4 shows, neither ensemble substantially outperforms the best individual models reported in Table 2 (LightGBM: F1 = 0.491), indicating that the classifiers converge on similar decision boundaries.

Additionally, NearMiss (v3) undersampling and SMOTEENN oversampling were evaluated as alternatives to cost-sensitive training. Undersampling degraded F1 and preci-

Ensemble	F1	Recall	Prec.	Acc.
Soft Voting	0.487	0.708	0.371	0.745
Hard Voting	0.492	0.640	0.400	0.775

Table 4: Ensemble CV performance (5-fold, training set).

sion consistently due to majority-class information loss. SMOTEENN showed no meaningful improvement either; the LightGBM model with SMOTEENN achieved F1 = 0.486 and recall = 0.714 on the test set, essentially matching the ensemble without resampling. Note that due to computational cost, SMOTEENN was evaluated with 2-fold CV only, making these results less reliable than the 5-fold estimates elsewhere. Overall, cost-sensitive learning via `scale_pos_weight` proves equally effective and substantially more computationally efficient.

5.2 Feature Importance

We conducted a SHAP (SHapley Additive exPlanations) analysis across all model families. Figure 2 shows the results for the LightGBM model. Across all models, the most influential predictors are consistently GenHealth, HighBP, BMI, AgeGroup, HighChol, and Comorbidity_Count—all established Type 2 diabetes risk factors (Centers for Disease Control and Prevention 2024). The rankings are stable across model families, confirming reliable signal capture.

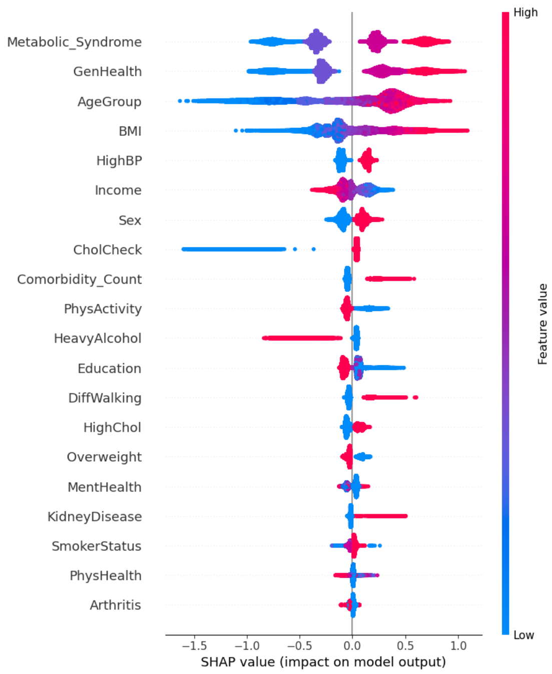

Figure 2: SHAP feature importance for the LightGBM model on the test set.

5.3 Final Test Set Performance

Table 5 reports the performance on the held-out test set (30%, n = 77,781). Results closely mirror cross-validation estimates, confirming generalizability. The soft voting ensemble achieves the best F1 (0.486) and AUC-ROC (0.808); the Balanced Random Forest yields the highest recall (0.770). For a screening application, the ensemble offers the best balance of recall and precision; where minimizing missed diagnoses is the absolute priority, the Balanced Random Forest is preferable.

Model	Acc.	Prec.	Recall	F1	AUC
Soft Voting Ensemble	0.743	0.369	0.713	0.486	0.808
Balanced Random Forest	0.704	0.338	0.770	0.470	0.804
LightGBM + SMOTEENN*	0.742	0.368	0.714	0.486	0.806

Table 5: Final test set performance (n = 77,781). *2-fold CV only.

Table 6 shows the confusion matrix for the soft voting ensemble, since it showed the best overall performance by F1 and AUC-ROC. Of 11,045 diabetic cases, 7,875 are correctly identified, while 3,170 are missed—meaning approximately 29% of diabetic individuals in the test set remain undetected, highlighting a limitation for standalone clinical use. The 13,464 false positives would be resolved through follow-up clinical testing.

		Predicted	
		No Diabetes	Diabetes
Actual	No Diabetes	53,272 (TN)	13,464 (FP)
	Diabetes	3,170 (FN)	7,875 (TP)

Table 6: Confusion matrix, soft voting ensemble (n = 77,781).

5.4 Comparison to State of the Art

Table 7 compares our results to published BRFSS-based studies. Such comparisons require caution, as the BRFSS survey instrument, variable definitions, sample compositions, and size change across years. Among studies using the same 2023 BRFSS data, we outperform Muhammad et al. (2025) on F1 and recall, likely due to our substantially larger sample (259,269 vs. 5,634) and additional feature engineering. Adewole et al. (2026) (BRFSS 2022) and Esmaeily et al. (2022) (earlier BRFSS waves) are less directly comparable due to differing survey years; Esmaeily et al. (2022) report a higher F1 (0.59), likely attributable to different target variable definitions. Studies using balanced test sets (Islam et al. 2025; Li et al. 2024) report much higher metrics but do not reflect real-world class distributions.

Study	Data	F1	Recall	AUC	Bal.?
Muhammad et al. (2025)	BRFSS ’23	0.37	0.32	0.80	No
Adewole et al. (2026)	BRFSS ’22	0.40	0.70	—	No
Esmaeily et al. (2022)	BRFSS	0.59	—	0.86	No
Islam et al. (2025)	BRFSS ’15	—	—	0.99	Yes
Li et al. (2024)	BRFSS ’21	0.95	—	0.99	Yes
This study (Ens.)	BRFSS ’23	0.49	0.71	0.81	No
This study (Bal. RF)	BRFSS ’23	0.47	0.77	0.80	No

Table 7: Comparison to published studies. “—” = not reported; “Bal.” = test set balanced.

6 Conclusion

We show that predicting diabetes risk from BRFSS 2023 survey data is feasible, with the best models achieving AUC-ROC $\approx$ 0.81 and F1 $\approx$ 0.49. Recall-oriented models identify up to 77% of diabetic individuals, suitable for screening where false negatives carry high cost. Cost-sensitive training proves as effective as resampling at lower computational cost. The low precision (0.34–0.41) reflects an inherent limitation of survey-based predictions and underscores the need for a confirmatory diagnostic step in practice. Key limitations include the reliance on self-reported health indicators rather than clinical biomarkers (e.g., HbA1c) and the cross-sectional survey design, which rules out causal inference.

References

Adewole, L. B., A. A. Aworeni, S. A. Mogaji, and T. Fele (2026). Addressing class imbalance in diabetes detection: A comparative study of SMOTE and under sampling techniques using BRFSS-2022 dataset. *Journal of Innovation Science and Technology*.

American Diabetes Association (2024). Standards of care in diabetes—2024. *Diabetes Care 47*(Supplement 1), S1–S350.

Centers for Disease Control and Prevention (2024). Type 2 diabetes risk factors. https://www.cdc.gov/diabetes/risk-factors/index.html.

Esmaeily, H., M. Tayefi, M. Ghayour-Mobarhan, and A. Amirabadizadeh (2022). Diabetes mellitus risk prediction in the presence of class imbalance using flexible machine learning methods. *BMC Medical Informatics and Decision Making 22*(36).

International Diabetes Federation (2025). IDF diabetes atlas. https://diabetesatlas.org/.

Islam, M. N., M. M. H. Rimon, S. S.-E.-A. Shamim, Z. M. Fahad, M. J. I. Mony, and M. J. U. Chowdhury (2025). An improved ensemble-based machine learning model with feature optimization for early diabetes prediction. arXiv:2512.02023.

Knowler, W. C., E. Barrett-Connor, S. E. Fowler, et al. (2002). Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *New England Journal of Medicine 346*(6), 393–403.

Li, Y. et al. (2024). Diabetes prediction model based on GA-XGBoost and stacking ensemble algorithm. *PLOS ONE 19*(9), e0311222.

Muhammad, M. A., J. Sani, and M. M. Ahmed (2025). Exploring explainable machine learning for predicting and interpreting self-reported diabetes among Tennessee adults: Insights from the 2023 BRFSS. *Journal of Primary Care & Community Health 16*.

National Health Service (2023). Type 2 diabetes — symptoms. https://www.nhs.uk/conditions/type-2-diabetes/symptoms/.

Rubin, R. R. and M. Peyrot (1999). Quality of life and diabetes. *Diabetes/Metabolism Research and Reviews 15*(3), 205–218.

Tuomilehto, J., J. Lindström, J. G. Eriksson, et al. (2001). Prevention of type 2 diabetes mellitus by changes in lifestyle among subjects with impaired glucose tolerance. *New England Journal of Medicine 344*(18), 1343–1350.

Contribution Overview

Manuel Dieterle contributed to the project proposal and was actively involved in discussions around model selection, the interpretation of cross-validation results, and the comparison to state-of-the-art studies. He authored Section 5 (Results), translating the experimental outputs into a coherent analytical narrative covering ensemble performance, feature importance, test set evaluation, and benchmarking against published work. He also conducted a thorough review of the entire report for technical consistency across sections and produced the final formatted version of the document.

Fatima Iqbal led the dataset identification process jointly with Özlem Ölçer, systematically evaluating over 300 BRFSS survey variables across multiple years to determine the most relevant feature subset. She contributed to feature engineering and performed the class balance and exploratory data analysis. In the report, she authored the sections on exploratory data analysis, modeling approaches, nested cross-validation setup, and evaluation metrics.

Tim Mauscherning researched and implemented the four clinically motivated composite proxy features — Metabolic Syndrome, High CVD Risk Profile, Comorbidity Count, and Healthy Lifestyle Score — drawing on medical literature to capture risk patterns not directly encoded in the survey. He authored Section 4 (Evaluation), covering the nested cross-validation results, hyperparameter optimization process, and robustness checks across model families.

Muhammad Murodzoda was responsible for the core technical implementation of the project. He built the full modeling pipeline, including the nested cross-validation framework, hyperparameter search via GridSearchCV, and metric stability analysis across training folds. He further implemented and evaluated ensemble methods (soft and hard voting), resampling strategies (SMOTEENN oversampling and NearMiss undersampling), and produced all final results on the held-out test set. His work forms the empirical backbone of the report.

Özlem Ölçer jointly led the dataset identification and variable selection with Fatima Iqbal, and was responsible for the data cleaning pipeline including duplicate removal, recoding of binary and ordinal variables, handling of invalid survey responses, and BMI rescaling. She authored Section 3.1 (Data Cleaning) and Section 3.2 (Feature Engineering), providing detailed documentation of all preprocessing decisions and the four engineered composite features with their formal definitions.

Xuemei Wang contributed to the project proposal and was involved in key methodological discussions throughout the project, including variable selection, target variable definition, and the choice of evaluation metrics in light of class imbalance. She contributed to the group's decision to prioritize recall as the primary evaluation metric, grounding this choice in the asymmetric costs of false negatives in a clinical screening context. She structured the overall report, authored Sections 1 (Application Area and Goals) and 2 (Dataset Profile), conducted a final proofreading pass across the full report, and drafted this contribution overview.

Ehrenwörtliche Erklärung

Wir versichern, dass wir die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Wir sind uns bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
Claude	Report writing assistance, structuring	Throughout	+
Claude	Literature search for state-of-the-art	Throughout	+
Gemini	Search for Dataset	Early on	-
Gemini	Code assistance	Notebook	+

Unterschrift
Mannheim, den 17. Mai 2026